

Intak Hwang

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I'm a Ph.D. student in [Cryptography & Privacy Lab](#) at Seoul National University, advised by [Yongsoo Song](#). I'm interested in post-quantum cryptographic protocols based on lattices, including but not limited to Fully Homomorphic Encryption and Zero-Knowledge Proofs.

Education

Seoul National University

2023 — Present

Integrated M.S. & Ph.D. in Computer Science & Engineering

Advised by Yongsoo Song

DGIST

2018 — 2022

B.S. in School of Undergraduate Studies

Summa Cum Laude

Conference Publications

[C10] [2025/395](#)

On the (In)security of Approximate Computation Protocols from CKKS

with Yisol Hwang, Miran Kim, Dongwon Lee, Yongsoo Song

Asiacrypt 2026

[C09] [2025/1804](#)

HERDS: Multi-Key Fully Homomorphic Encryption with Sublinear Bootstrapping

with Binwu Xiang, Seonhong Min, Zhiwei Wang, Haoqi He, Yuanju Wei, Kang Yang, Jiang Zhang, Yi Deng, Yu Yu

Eurocrypt 2026

[C08] [2024/2032](#)

Carousel: Fully Homomorphic Encryption with Bootstrapping over Automorphism Group

with Seonhong Min, Yongsoo Song

Asiacrypt 2025

[C07] [2025/382](#)

On the Security and Privacy of CKKS-based Homomorphic Evaluation Protocols

with Seonhong Min, Jinyeong Seo, Yongsoo Song

Asiacrypt 2025

[C06] [2025/216](#)

Practical TFHE Ciphertext Sanitization for Oblivious Circuit Evaluation

with Jinyeong Seo, Seonhong Min, Yongsoo Song

ACM CCS 2025

[C05] [2024/1879](#)

Practical Zero-Knowledge PIOP for Maliciously Secure Multiparty Homomorphic Encryption

with Hyeonbum Lee, Jinyeong Seo, Yongsoo Song

ACM CCS 2025

[C04] [2025/1255](#)

Efficient Full Domain Functional Bootstrapping from Recursive LUT Decomposition

with Shinwon Lee, Seonhong Min, Yongsoo Song

SAC 2025

[C03] [2024/1502](#)

MatriGear: Accelerating Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing

with Hyunho Cha, Seonhong Min, Jinyeong Seo, Yongsoo Song

IEEE S&P 2025

[C02] [2024/306](#)

Concretely Efficient Lattice-based Polynomial Commitment from Standard Assumptions

with Jinyeong Seo, Yongsoo Song

Crypto 2024

[C01] [2023/1328](#)

Optimizing HE Operations via Level-aware Key-switching Framework

with Jinyeong Seo, Yongsoo Song

WAHC 2023

Journal Publications

[J01]

A Privacy-Preserving HLA Imputation Method with Homomorphic Encryption

with Hakin Kim, Yongsoo Song, Buhm Han

iScience

Preprints

[P03] [2026/2120](#)

Multi-Key FHE with Compact Ciphertexts (or: MKFHE Strikes Back)

with Seonhong Min, Yongsoo Song

[P02] [2026/044](#)

Jindo: Practical Lattice-Based Polynomial Commitments for Client-Side Proving

with Hyeonbum Lee, Jinyeong Seo, Yongsoo Song

[P01] [2025/203](#)

Ciphertext-Simulatable HE from BFV with Randomized Evaluation

with Seonhong Min, Yongsoo Song

Presentations

Practical Zero-Knowledge PIOP for Maliciously Secure Multiparty Homomorphic Encryption

FHE.org 2026 — Taipei, Taiwan

Practical TFHE Ciphertext Sanitization for Oblivious Circuit Evaluation

ACM CCS 2025 — Taipei, Taiwan

FHE.org — [Online](#)

MatriGear: Accelerating Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing

IEEE S&P 2025 — San Francisco, USA

Optimizing HE via Level-aware Key-switching

WAHC 2023 — Copenhagen, Denmark

Projects

TFHE-go

TFHE-go is an implementation of the (MK)TFHE scheme, written in Go and Go assembly. It is currently one of the fastest and most feature-complete open-source TFHE implementations available.

Ringo-SNARK

Ringo-SNARK is a Zero-Knowledge PIOP toolkit for efficiently proving Ring-LWE relations, written in Go. It supports simple, gnark-like circuit design and compilation.

Honors and Scholarships

National Cryptographic Contest

Excellence Award, Encouragement Award	2025
Best Award, Excellence Award	2024
Special Award	2023

CTF Security Competitions

	2020 — 2022
SSTF Hacker's Playground 2022	5th place
WhiteHat Contest 2021	3rd place
DEFCON CTF 2021	Finalist
PlaidCTF 2021	5th place
Real World CTF 2020/2021	1st place
Midnight Sun CTF 2020 Finals	7th place

TokyoWesterns CTF 2020 Finals

3rd place

DEFCON CTF 2020

Finalist

DGIST Dean's List

2020

Skills

Languages

Korean (native), English (fluent)

Programming Languages

Go, Python (SageMath), C/C++, Rust, LaTeX

Other Activities

Member of CTF Team CodeRed

2020 — 2022

I participated in CTF competitions from time to time, mostly solving crypto challenges.